Regularised methods

Oxford Spring School in Advanced Research Methods

Dr Thomas Robinson, LSE

Day 2 of 5

Introduction

Yesterday we explored how a familiar estimator (logistic regression) incorporates some fundamental aspects of ML

- ML is not some entirely new, alien type of doing statistics
- ML is typically focused on prediction problems
- Lots of really useful ML models are extensions of regression framework

So how should we understand OLS within a prediction context?

How do other popular forms of ML come out of it?

Today's session

1. Recap OLS from prediction perspective
    - How does OLS work?
    - Optimisation criteria
    - Bias-variance trade-off
2. LASSO estimator
3. Practical application

Key topics:

- Bias and variance
- Regularisation
- Validation and leakage

Running case study lens

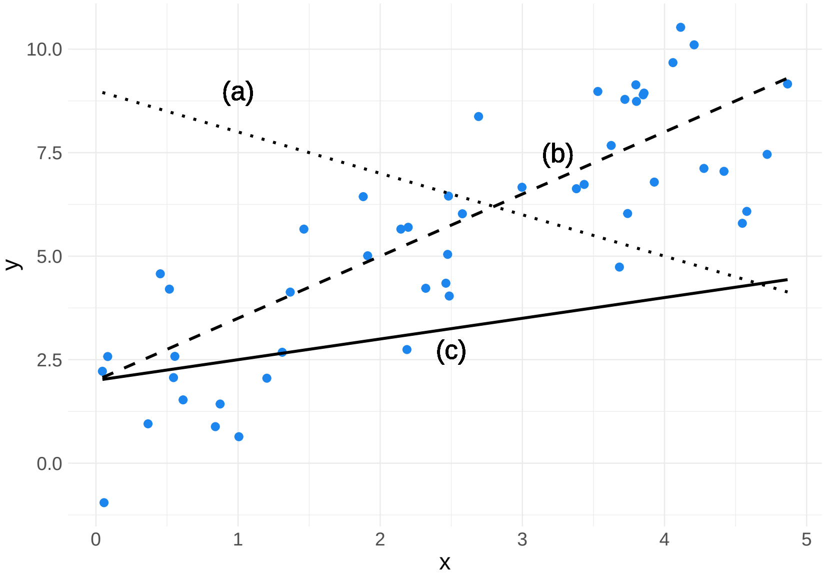
Keep the week's survey prediction case in mind:

- Many plausible controls
- Many possible interactions
- High risk of overfitting if we keep expanding the feature set

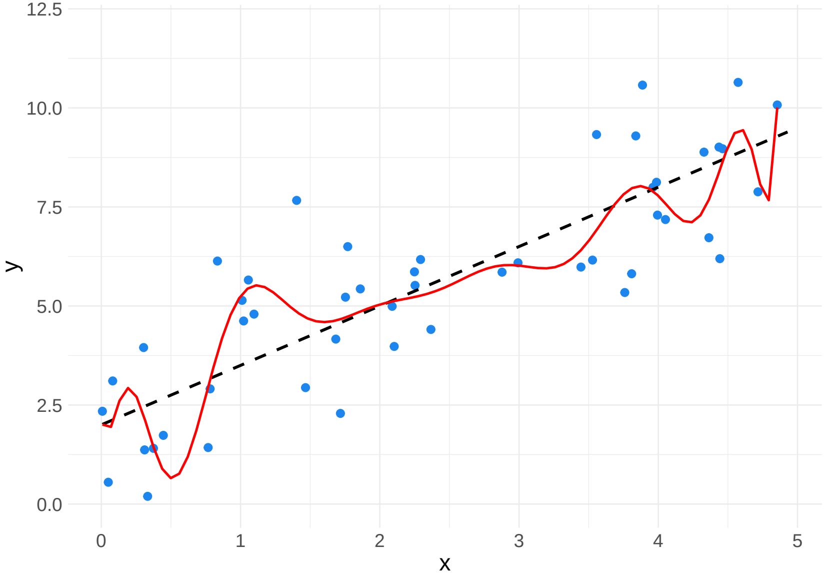
Regularisation helps when the research problem produces more candidate predictors than we can defend one-by-one.

Ordinary Least Squares Regression

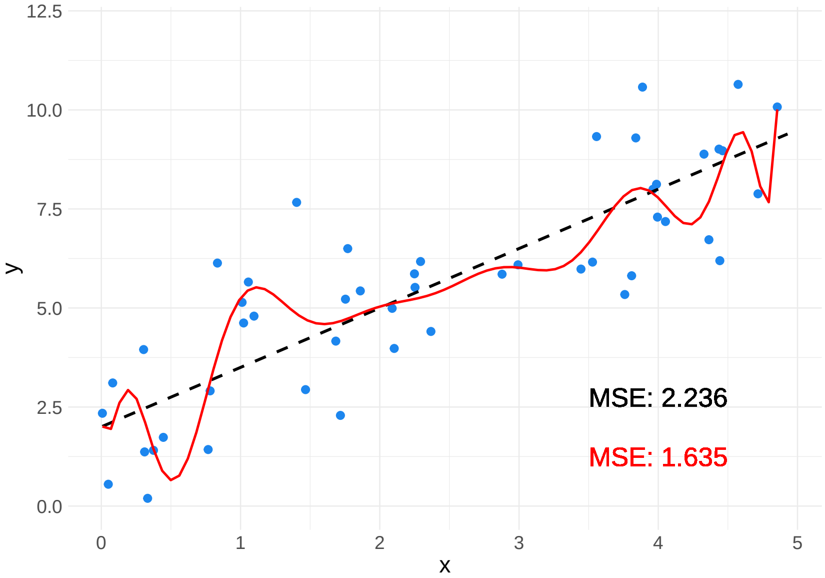
Refresher: which is the best line?



Refresher: which is the best line? 2



Refresher: comparing model fits by MSE



OLS as a tool for inference

In inference terms, OLS estimates $\hat{\beta}$ that:

- Capture the linear relationships between $\mathbf{X}$ and $\mathbf{y}$
- Yield individual estimates of the partial associations between $\boldsymbol{X}$ on $\boldsymbol{y}$
- Allows us to understand the uncertainty over $\hat{\boldsymbol{\beta}}$
  - E.g., how confident we are that there is a non-zero relationship between $\boldsymbol{x_1}$ and $\boldsymbol{y}$

OLS: Optimisation

OLS regression minimises the **sum of the squared error** between the regression line ($\boldsymbol{X}\beta$) and the observed outcome ($\mathbf{y}$):

$$\arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{N} (y_i - \boldsymbol{x_i}\boldsymbol{\beta})^2,$$

where $\boldsymbol{x_i}\boldsymbol{\beta}$ is the linear regression function.

How might we solve this?

- Maximum likelihood estimation
- Calculus – there is a closed form solution (unlike logistic regression)

Why do we like OLS?

Not only does OLS have a closed form solution, we also know that, under the Gauss Markov (GM) assumptions, OLS is:

- **B**est
- ~~**L**inear~~ (Hansen 2021)
- **U**nbiased
- **E**stimator

GM Assumptions

In other words, in terms of estimating the parameters β, you will not find a model with a lower variance that is also unbiased

OLS is good for inference:

- Typically very concerned with generating **unbiased** estimates of $\hat{\beta}$
- Most efficient test of a (linear) hypothesis

Bias

Bias is a feature of the estimator:

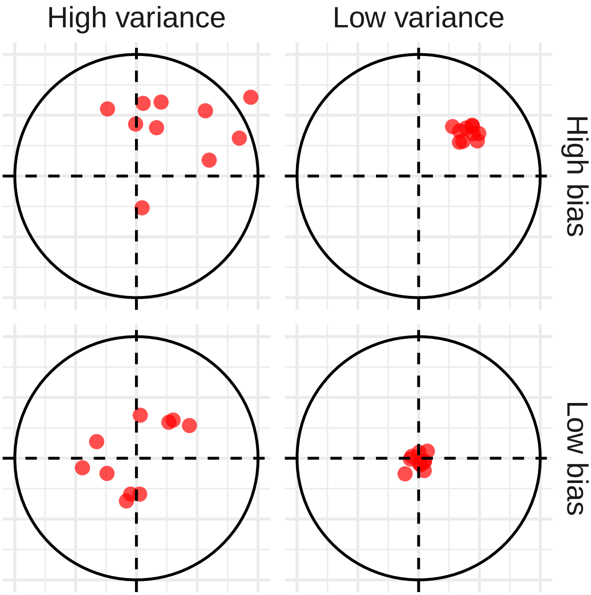
- $\text{Bias}_{\boldsymbol{\beta}} = (\mathbb{E}[\hat{\boldsymbol{\beta}}] - \boldsymbol{\beta})$
- On average, the estimated parameters are equal to the true parameters
- Under GM, we know that $(\mathbb{E}[\hat{\boldsymbol{\beta}}] - \boldsymbol{\beta}) = 0$

Variance

As we sample new (GM-satisfying) data, the parameters of our model will shift:

- Hence, there will be variance over our parameter estimates
- $\mathbb{V}_{\hat{\boldsymbol{\beta}}} = \mathbb{E}\big[(\mathbb{E}[\hat{\boldsymbol{\beta}}] - \hat{\boldsymbol{\beta}})^2\big]$
- The average distance between a particular parameter estimate and the mean of parameter estimates over multiple samples

Visualising bias and variance



Predicting *new* values

In the remainder of today's session, we are going to consider the following generic supervised learning problem:

- We observe $(\boldsymbol{y}, \boldsymbol{X}) \in \mathcal{D}$
  - A training sample that is taken from a wider possible set of data
  - I.e. we can think, counterfactually, of resampling to get a new sample $(\boldsymbol{y}_{\text{New}}, \boldsymbol{X}_{\text{New}})$
- We also observe a "test" dataset $\boldsymbol{X'}$

The goal is to estimate $\boldsymbol{y'}$ by training a model $\hat{f}$

- The outcomes that correspond to $\boldsymbol{X'}$

OLS as a tool for prediction

When we run OLS, we also get a “trained model”:

- $\hat{f}$ – that has parameters equal to $\hat{\beta}$
- Can be applied to a new “test” dataset $\boldsymbol{X'}$
- To generate new predictions $\boldsymbol{y'}$

Train, validation, and test data

For predictive work, keep three roles separate:

- Training data
  - Used to estimate model parameters
- Validation / resampling data
  - Used to choose hyperparameters such as λ
- Test data
  - Used once, at the end, for an honest estimate of performance

If we tune on the test set, it stops being a test set.

Bias and variance of predictions

We can also think of bias in terms of the predictions:

- $\text{Bias}_{\boldsymbol{y}} = (\mathbb{E}[\hat{\boldsymbol{y}}] - \boldsymbol{y})$
- We ideally want low bias
- High bias suggests the model is not sensitive enough

And we can think about the variance of the prediction:

- $\mathbb{V}_{\hat{\boldsymbol{y}}} = \mathbb{E}\big[(\mathbb{E}[\hat{\boldsymbol{y}}] - \hat{\boldsymbol{y}})^2\big]$

High variance means that the model is very sensitive to $\boldsymbol{X}$ – the training data – but will perform poorly out-of-sample

- With new data, and high variance, we would expect quite different predictions

Bias-variance trade off

So can't we just choose a low-variance, low-bias modeling strategy?

Assume we could calculate the mean squared error of some test data $\boldsymbol{X'}$ given a trained model $\hat{f}$:

$$\text{MSE} = \mathbb{E}[(\hat{f}(\boldsymbol{X'}) - \boldsymbol{y'})^2].$$

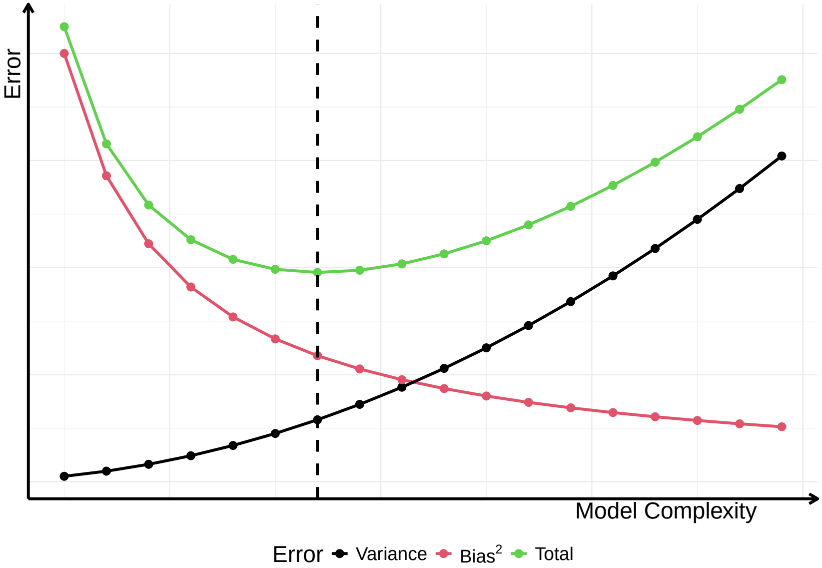
We can decompose this further:

$$MSE = \underbrace{\mathbb{E}\big[(\hat{f}(\boldsymbol{X'}) - \mathbb{E}[\hat{\boldsymbol{y}}])^2\big]}_{\text{Variance}} + \underbrace{\big(\mathbb{E}[\hat{\boldsymbol{y}}] - \boldsymbol{y'}\big)^2}_{\text{Bias}^2}$$

So holding the MSE fixed, if we reduce the variance we must increase the bias

- I.e. there is a **bias-variance trade-off**

Visualising the trade-off

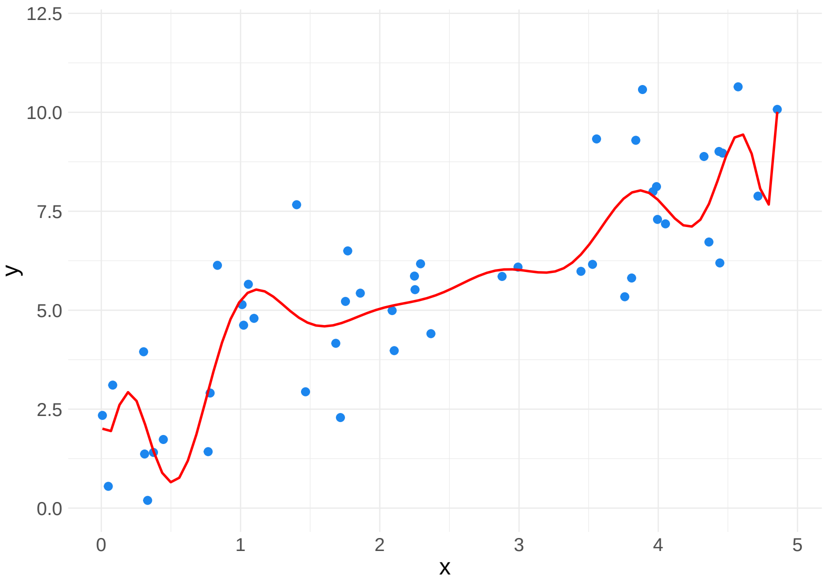


Out of sample performance of OLS

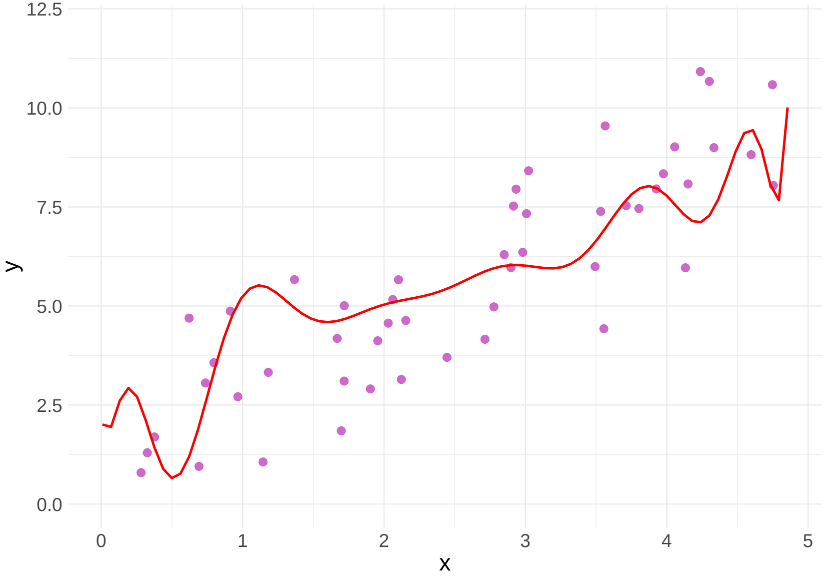
This trade-off explains why we might not want to use OLS for prediction tasks:

- By virtue of GM assumptions and B(L)UE, the MSE is explained entirely by the variance
  - Across models, the parameters will be centred on the true values (an unbiased estimator)
- We cannot tweak the model to get slightly better out-of-sample predictions at the expense of some added bias
  - In other words, we cannot leverage the bias-variance trade-off

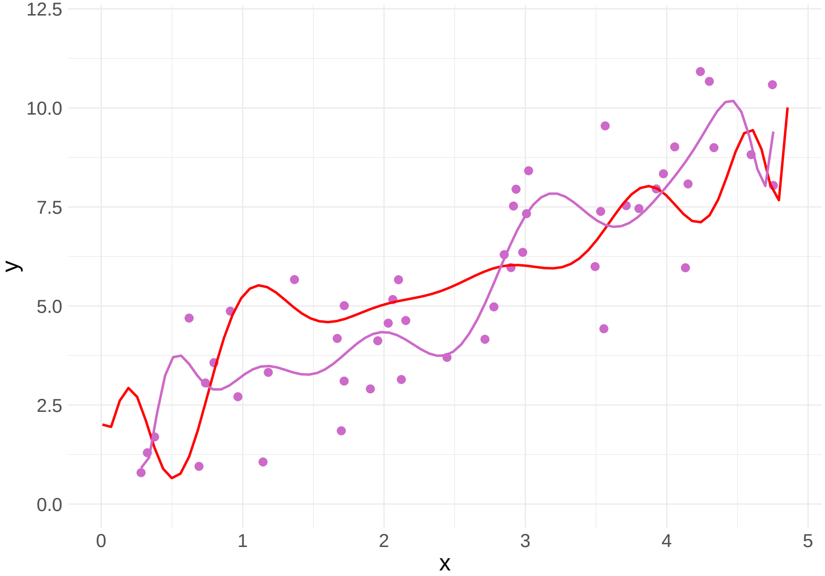
OLS: Complex OLS model trained on $\mathbf{X}$



OLS: Compare models' predictions to $\mathbf{X}'$



OLS: Variance in model predictions over $\mathbf{X}$



The LASSO estimator

Regularisation and overfitting

In the previous examples, a complex model yields poor out-of-sample predictions.

To ensure our model does not have too much variance, we can *penalise* this complexity

- This will introduce bias into the model
- But, if done well, we can reduce the *total* MSE by offsetting overly-high variance
- And therefore yield better predictions on $\boldsymbol{X'}$

We call this process **regularisation**:

- Especially important where, by design, ML algorithms are highly flexible/complex
- Limits **overfitting**

Regularisation of OLS

If we want to regularise OLS we need to modify its loss function L:

- L_{OLS} is unbiased
- So, adding any non-zero term will *add* bias...
- ... and hopefully improve out-of-sample prediction

In other words:

- We sacrifice some variance in order to improve the predictive performance of the model on $\boldsymbol{X}'$

Generalising OLS with regularization

Following Kleinberg et al (2015) we can state this problem as the linear optimisation of:

$$\arg\min_f \underbrace{\sum_{i=1}^{N}(y_i - f(\boldsymbol{x_i}))^2}_{\text{Sum of squared error}} + \underbrace{\lambda R(f)}_{\text{Regularisation}}$$

For OLS:

- $f \in \mathcal{F}_{\text{linear}}$
- $\lambda = \frac{1}{\infty} = 0$

But what if $\lambda \neq 0$?

- Then we must decide what $R(\cdot)$ is
- And decide on a value of λ – a hyperparameter (we'll use cross-validation in the coding session, and revisit tuning on Day 3)

$R(\cdot)$ as shrinkage

Consider an OLS model with k parameters:

- The model estimates coefficients for each parameter
- Regardless of how large or small that coefficient is
- In a sense, with non-zero estimates for each parameter these models can be considered "complex"

We can reduce the complexity of the regression model by setting some parameters to zero

- I.e. we **shrink** the coefficient estimates
- Aim to reduce the variance error by more than the increase in bias error

In the linear framework, we need some way to penalize non-zero coefficients

Least absolute shrinkage and selection operator (LASSO)

We can calculate the total magnitude (or **L1 Norm**) of the coefficients in a model as:

$$||\boldsymbol{\beta}||_1 = \sum_j |\beta_j|$$

Next, we can think about restricting the size of this norm:

$$||\boldsymbol{\beta}||_1 \leq t$$

Finally we can include this term in our loss function:

$$\arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{N} (y_i - \boldsymbol{x_i}\boldsymbol{\beta})^2 \text{ subject to } ||\boldsymbol{\beta}||_1 \leq t,$$

which is equivalent to:

$$\arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{N} (y_i - \boldsymbol{x_i}\boldsymbol{\beta})^2 + \lambda ||\boldsymbol{\beta}||_1.$$

LASSO

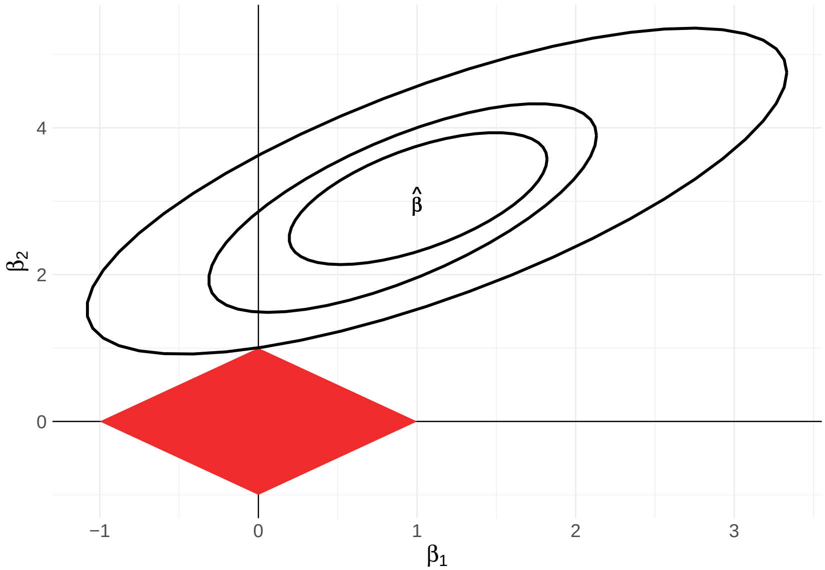
Hence, the LASSO estimator conforms to the generalised loss function introduced earlier

- $R(f) = ||\hat{\boldsymbol{\beta}}||_1$

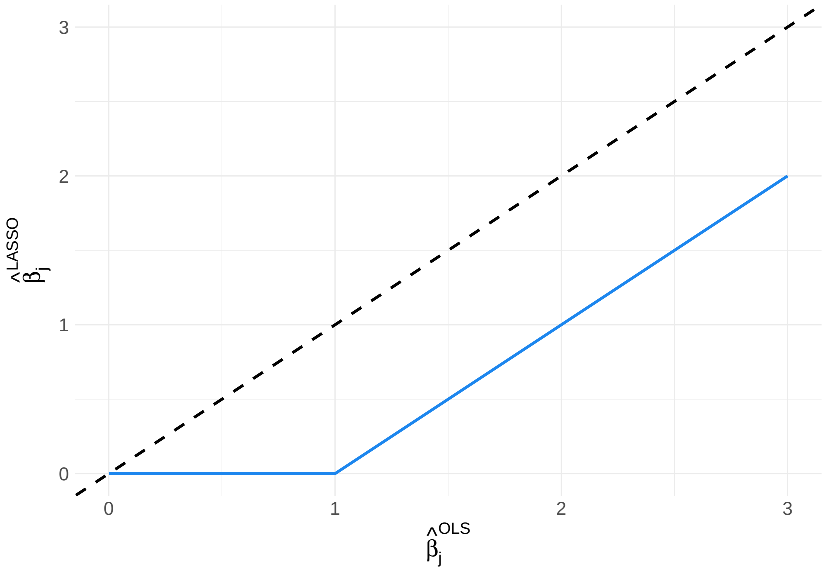
LASSO often yields coefficient estimates of exactly zero:

- Think about varying the true value of some coefficient β_j:
  - When β_j is large, we might shrink it (relative to OLS) but the importance of this predictor is sufficient to entail a non-zero coefficient
  - But for some small enough value b, the cost of including b in L1-norm is greater than the reduction in squared error
- In other words, the L1 norm constraint can lead to "corner solutions"

Example of LASSO corner solution ($||\beta||_1 \leq 1$)



Comparison of $\hat{\beta}_j^{\text{OLS}}$ to $\hat{\beta}_j^{\text{LASSO}}$



Two helpful properties of LASSO

1. Prediction accuracy
   - We trade off an amount of bias for a (hopefully) greater reduction in variance, improving out-of-sample prediction
   - Cf. a non-zero *true* coefficient estimated with a large confidence interval
2. Selection of relevant variables
   - The possibility of corner-solutions acts as a useful variable selection mechanism
   - LASSO essentially selects the most important variables for us

When LASSO is especially useful

LASSO is often a strong default when:

- The researcher has many controls, interactions, or fixed effects
- p is not small relative to n
- Prediction or nuisance adjustment matters more than interpreting every coefficient
- The goal is to reduce a very large candidate feature set before a second-stage analysis

If predictors are strongly correlated, elastic net (which combines L1 and L2 penalties) is often worth considering too.

Application

Generalised applications

What sorts of general topics/data might we expect to see LASSO models?

Generalised applications

What sorts of general topics/data might we expect to see LASSO models?

- Mueller Hannes, Rauh Christopher. Reading Between the Lines: Prediction of Political Violence Using Newspaper Text. American Political Science Review. 2018;112(2):358-375.
  - Topic selection
- Kim In Song. Political Cleavages within Industry: Firm-level Lobbying for Trade Liberalization. American Political Science Review. 2017;111(1):1-20.
  - Word selection
- Gerring John, Jerzak Connort, Öncel Erzen. The Composition of Descriptive Representation. American Political Science Review. 2023:1-18.
  - Variable selection more generally

Blackwell and Olson (2021)

Blackwell, Matthew and Michael P. Olson. "Reducing Model Misspecification and Bias in the Estimation of Interactions." *Political Analysis*, 2021.

Suppose we have an outcome $\boldsymbol{y}$, a treatment $\boldsymbol{d}$, covariates $\boldsymbol{X}$, and an "effect moderator" $\boldsymbol{v}$

- We want to estimate an *inference* model
- Understand how the treatment effect is moderated

Naive suggestion:

- Include a single interaction term
  - I.e. $y_i = \beta_0 + \beta_1 d_i + \beta_2 v_i + \beta_3 d_i v_i + \boldsymbol{\beta' X_i}$

What could be wrong with this model?

Blackwell and Olson (2021)

Blackwell, Matthew and Michael P. Olson. "Reducing Model Misspecification and Bias in the Estimation of Interactions." *Political Analysis*, 2021.

Suppose we have an outcome $\boldsymbol{y}$, a treatment $\boldsymbol{d}$, covariates $\boldsymbol{X}$, and an "effect moderator" $\boldsymbol{v}$

- We want to estimate an *inference* model
- Understand how the treatment effect is moderated

Naive suggestion:

- Include a single interaction term
  - I.e. $y_i = \beta_0 + \beta_1 d_i + \beta_2 v_i + \beta_3 d_i v_i + \boldsymbol{\beta' X_i}$

What could be wrong with this model?

- It implicitly assumes no interactions between $\boldsymbol{v}$ and $\boldsymbol{X}$ matter
- If the relationship between covariates and the outcome differs across levels of $\boldsymbol{v}$, omitting those $\boldsymbol{vX}$ interactions biases $\hat{\beta}_3$
- Blackwell and Olson call this **omitted interaction bias**

Why not just use LASSO directly?

Standard adaptive ML methods (adaptive LASSO, KRLS, BART) suffer from two biases:

- **Direct regularisation bias**: LASSO shrinks *all* coefficients, including the interaction of interest $\hat{\beta}_3$, toward zero
- **Indirect regularisation bias**: a LASSO on $\boldsymbol{y}$ alone may drop covariates that weakly predict $\boldsymbol{y}$ but strongly predict $\boldsymbol{d}$ or $\boldsymbol{dv}$ – causing omitted variable bias in the interaction estimate

The "double" in post-double selection addresses the second problem by running LASSO on the treatment side too

Combining LASSO and unpenalised regression

Blackwell and Olson propose splitting the problem into two stages:

1. **Variable selection** (LASSO)
   - Use LASSO as a *selection* device only – not for estimation
   - Run separate models for outcome, treatment, and treatment-moderator interaction
   - This avoids indirect regularisation bias
2. **Inference** (unpenalised regression)
   - Estimate an unpenalised model (OLS or logistic) using only the selected variables
   - This avoids direct regularisation bias

What makes this strategy so useful is that:

- LASSO handles the high-dimensional selection in Stage 1
- Unpenalised estimation in Stage 2 gives unbiased coefficients and valid standard errors

Post-double selection method

Stage 1

- Let $\boldsymbol{Z} = \{\boldsymbol{v}, \boldsymbol{X}, \boldsymbol{vX}\}$ be the candidate controls from a fully moderated model
- Estimate LASSO models for:
  1. $\boldsymbol{y}$ on $\boldsymbol{Z}$
  2. $\boldsymbol{d}$ on $\boldsymbol{Z}$
  3. $\boldsymbol{dv}$ on $\boldsymbol{Z}$
- Let $\boldsymbol{Z}^*$ index all variables with non-zero coefficients in any of models 1–3

Stage 2

- Regress $\boldsymbol{y}$ on $\boldsymbol{d}, \boldsymbol{dv}$ and $\boldsymbol{Z}^*$ using an unpenalised model

Blackwell and Olson also force all "base-terms" ($\boldsymbol{v}$ and $\boldsymbol{X}$) into Stage 2 regardless of LASSO selection

Readings

Suggested readings after today's class:

- James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning*
  - Ch. 6 for regularisation and model selection
- Mullainathan and Spiess (2017), "Machine Learning: An Applied Econometric Approach"
  - Section on regularisation and the bias-variance trade-off
- Blackwell and Olson (2021), "Reducing Model Misspecification and Bias in the Estimation of Interactions"
  - The post-double-selection approach used in today's exercise
- Hastie, Tibshirani, and Friedman, *The Elements of Statistical Learning*
  - Ch. 3.4 for shrinkage methods and elastic net

Coding workshop: Implementing post-double selection using LASSO

Extra Slides

Alternative $R(f)$ to the L1 norm

Following a similar logic to the shrinkage used by LASSO, we can define other measures of magnitude, like the L2 norm $||\boldsymbol{\beta}||_2$:

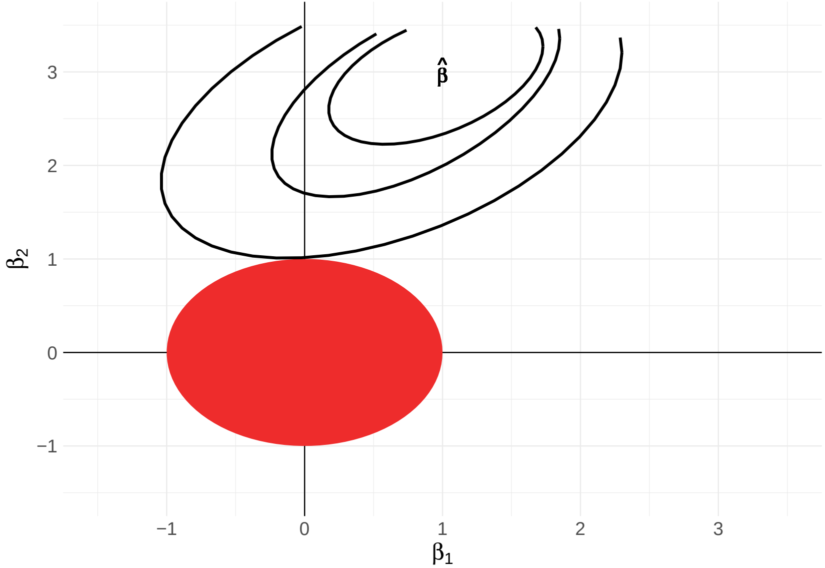
$$\sqrt{\sum_j |\beta_j|^2}$$

When we plug in the L2 norm into the loss function, we get the **ridge regression** estimator:

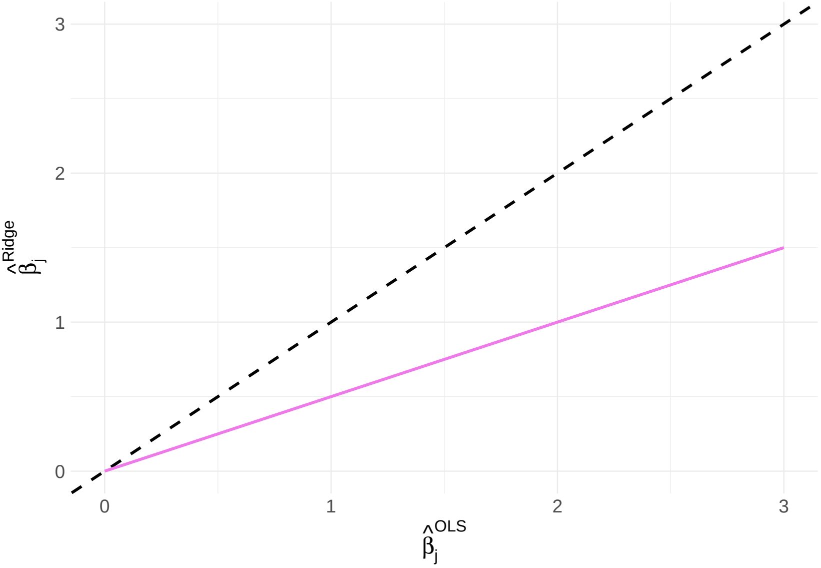
$$\arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{N} (y_i - \boldsymbol{x_i}\boldsymbol{\beta})^2 + \lambda||\boldsymbol{\beta}||_2$$

Unlike the LASSO estimator, ridge regression does not have a sharp cut-off, but rather scales the size of all coefficients in the model

Ridge regression – no corner solutions



Ridge regression – constant scaling of coefficients



Gauss Markov Assumptions

Five assumptions need to hold:

1. $\boldsymbol{y}$ is a linear function of $\boldsymbol{\beta}$
2. $\mathbb{E}[\epsilon_i] = 0$
3. $\mathbb{V}[\epsilon_i] = \sigma_i^2, \forall i$
4. $Cov(\epsilon_i, \epsilon_j) = 0, \forall i \neq j$
5. $Cov(\boldsymbol{x_i}, \epsilon_i) = 0$